Evolution of Real-Time Object Detection: From YOLOv1 to Modern Detectors

Real-time object detection models must solve two simultaneous tasks: identifying what objects are present in an image (classification) and predicting bounding box spatial coordinates (localization). Early approaches used two-stage detectors like R-CNN and Faster R-CNN, generating sparse region proposals followed by convolutional classification, which limited frame rates.

The You Only Look Once (YOLO) architecture revolutionized computer vision by framing detection as a single regression problem directly from image pixels to bounding box coordinates and class probabilities. By dividing the input image into an S-by-S grid and computing predictions in a single forward pass, YOLO achieved high frame rates suitable for real-time video feeds. Later iterations introduced anchor boxes, feature pyramid networks (FPN), path aggregation networks (PANet), and mosaic data augmentations.